

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5212
Subject Title	Engineering Computations and Modelling
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This course aims to equip students with essential skills and knowledge in computational methods and modeling techniques relevant to engineering practice. Through the course, students will learn fundamental computational algorithms and programming languages, gaining proficiency in numerical analysis, data visualization, and mathematical modeling. The course eventually seeks to prepare students to tackle complex engineering challenges using computational approaches and to foster a mindset conducive to continuous learning and innovation in the field.
Intended Learning Outcomes	Upon completion of this subject, students will: - Demonstrate proficiency in using computational tools and software packages for engineering analysis and modelling. - Apply numerical methods to solve engineering problems accurately and efficiently. - Develop mathematical models to represent and analyse engineering systems. - Recognize the importance of and exhibit the ability to engage in lifelong learning endeavours.
Subject Synopsis/ Indicative Syllabus	Introduction to Computational Tools: Overview of computational software packages such as MATLAB, Python, and C++; Basic programming concepts and syntax. Numerical Methods: Root finding methods (e.g., bisection, Newton-Raphson); Numerical integration and differentiation techniques; Solving systems of linear equations. Curve Fitting and Interpolation: Least squares fitting; Interpolation methods (e.g., Lagrange interpolation, spline interpolation). Differential Equations and Modelling: Introduction to ordinary and partial differential equations; Modelling physical systems using differential equations; Numerical solution methods for differential equations (e.g., Euler's method, Runge-Kutta methods).

	Fundamentals of Finite Element Analysis (FEA): Basics of finite element method; Mesh generation and discretisation; Solving linear and nonlinear FEA problems. Fundamentals of Computational Fluid Dynamics (CFD): Introduction to computational fluid dynamics; Governing equations of fluid flow; Numerical solution methods for CFD simulations. Optimisation Techniques: Introduction to optimisation problems; Gradient-based and gradient-free optimisation algorithms; Application of optimisation techniques in engineering design and analysis. Statistical Analysis and Data Visualisation: Descriptive statistics and probability distributions; Hypothesis testing and regression analysis; Data visualisation techniques.																																		
Teaching/Learning Methodology	The teaching and learning methods include lectures, mini-project and homework assignments. Technical/practical examples and problems will be raised and discussed in class. Project is designed to model engineering problems using computational skills. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Mini-project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Homework assignments</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	✓	✓	✓	✓	Mini-project	✓	✓	✓	✓	Homework assignments	✓	✓	✓	✓										
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Homework assignments</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Mini-project</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Final examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 0.5 × Continuous Assessment + 0.5 × Final Examination Continuous assessment, comprising homework assignments and project report, functions to track students' advancement, facilitate self-evaluation against intended learning goals, and foster the amalgamation of knowledge gained from lectures. Conversely, the	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Homework assignments	25%	✓	✓	✓	✓	2. Mini-project	25%	✓	✓	✓	✓	3. Final examination	50%	✓	✓	✓		Total	100%				
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2. Mini-project	25%	✓	✓	✓	✓																														
3. Final examination	50%	✓	✓	✓																															
Total	100%																																		

	final examination objectively assesses students' knowledge acquisition, capacity to understand and critically analyze problems independently, and overall attainment of subject learning objectives.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	39 Hrs.
	Other student study effort:	
	▪ Self-study	78 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Engineering Computations and Modeling in MATLAB®/Simulink®, 2nd Edition, 2019, O. Yakimenko, AIAA. 2. Simulation with Python: Develop Simulation and Modeling in Natural Sciences, Engineering, and Social Sciences, 1st Edition, 2022, A. Nakano and R. Li, Springer. 3. Essential Python for the Physicist, 1st Edition, 2020, G. Moruzzi, Springer. 4. Effective Computation in Physics: Field Guide to Research with Python, 1st Edition, 2015, A. Scopatz & K. Huff, O'Reilly.	

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